Nutrition Agent- using IBM Watsonx with IBM BOB

Create a new AI Agent – a Personalized Nutrition Agent that collects user preferences such as age, food preferences, health and medical conditions, and location (city, state, country), and suggests a suitable diet plan**.**

Description

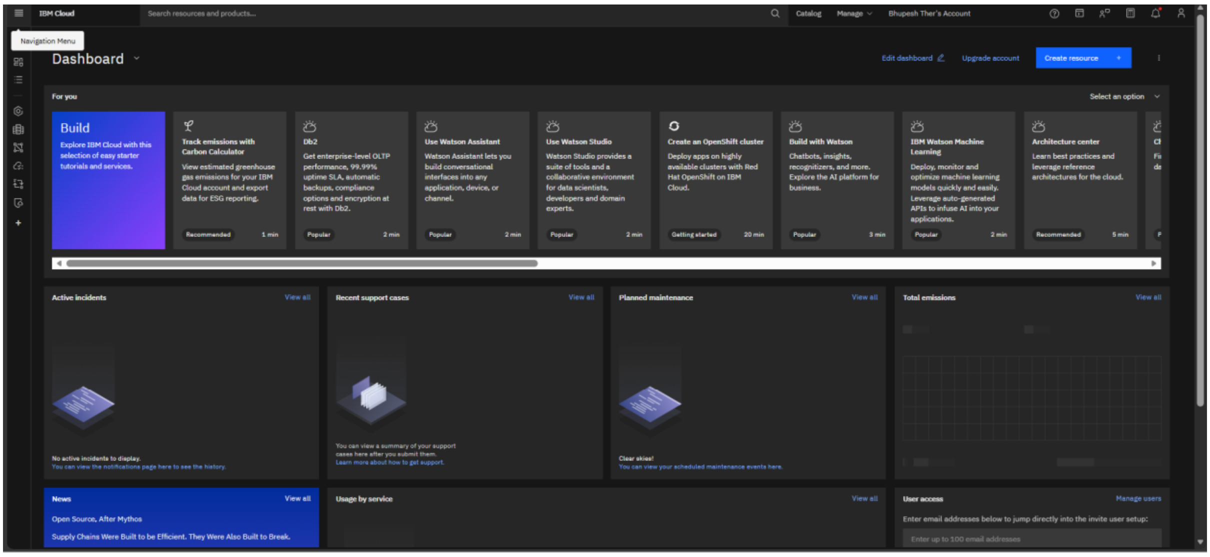
AI Agent – a Personalized Nutrition Agent that collects user preferences such as age, food preferences, health and medical conditions, and location (city, state, country), and suggests a suitable diet plan

Tools & Technologies Used:

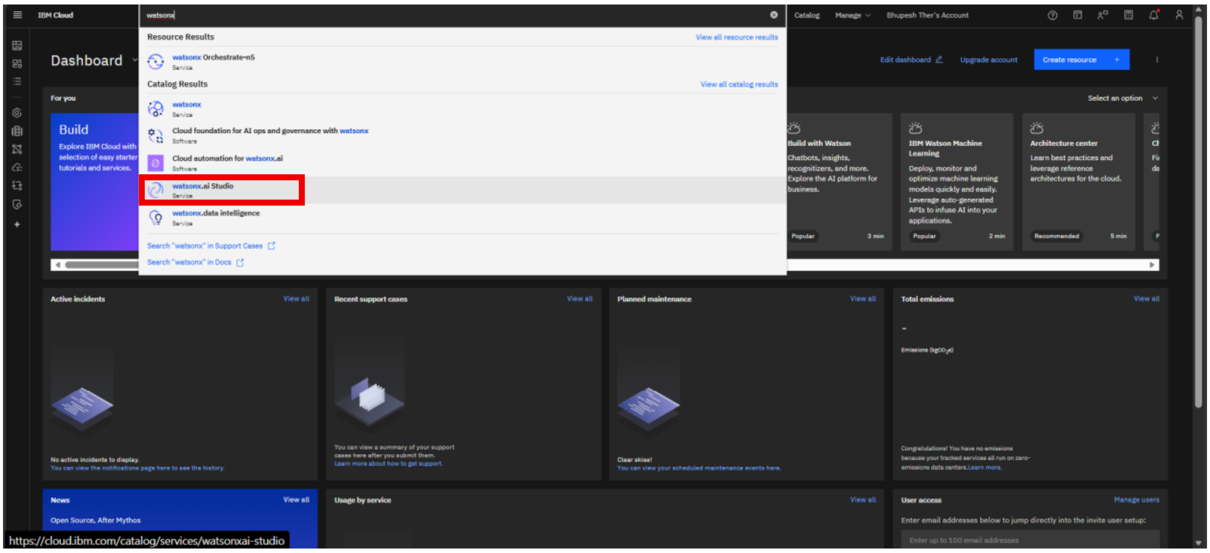
- IBM watsonx

-IBM BOB

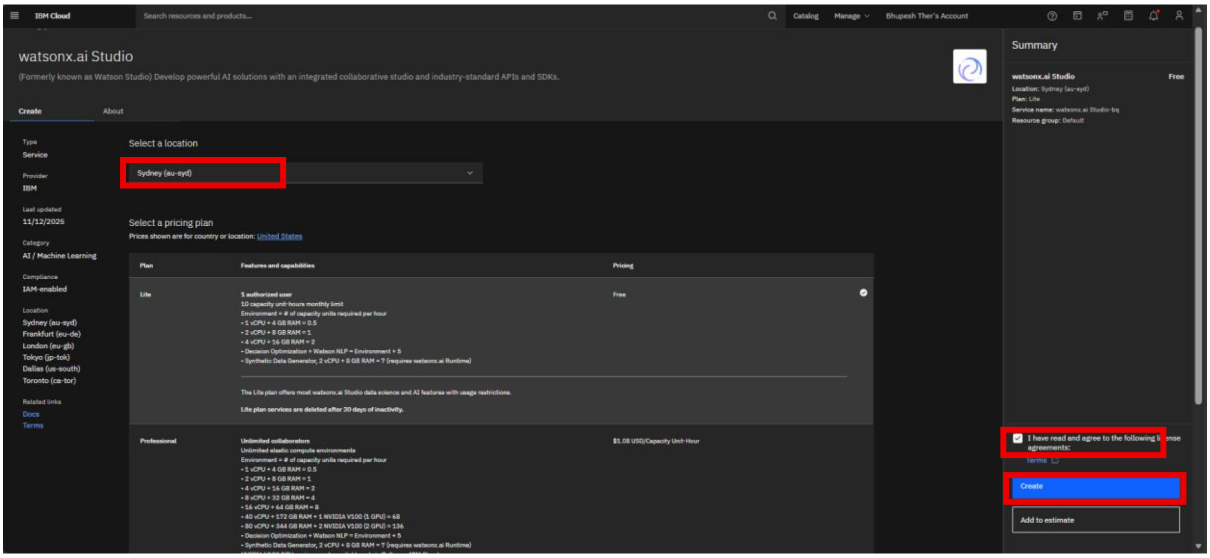
Step-1 login with your credential on IBM cloud.



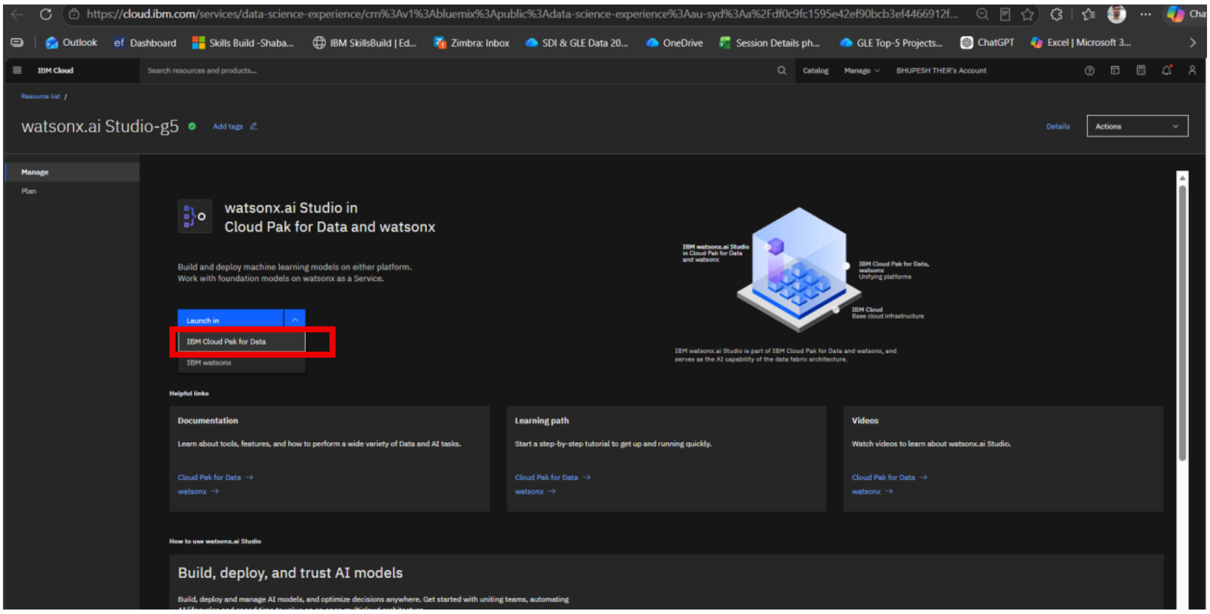
Step-2 go to search bar and search **watsonx.ai.studio** .



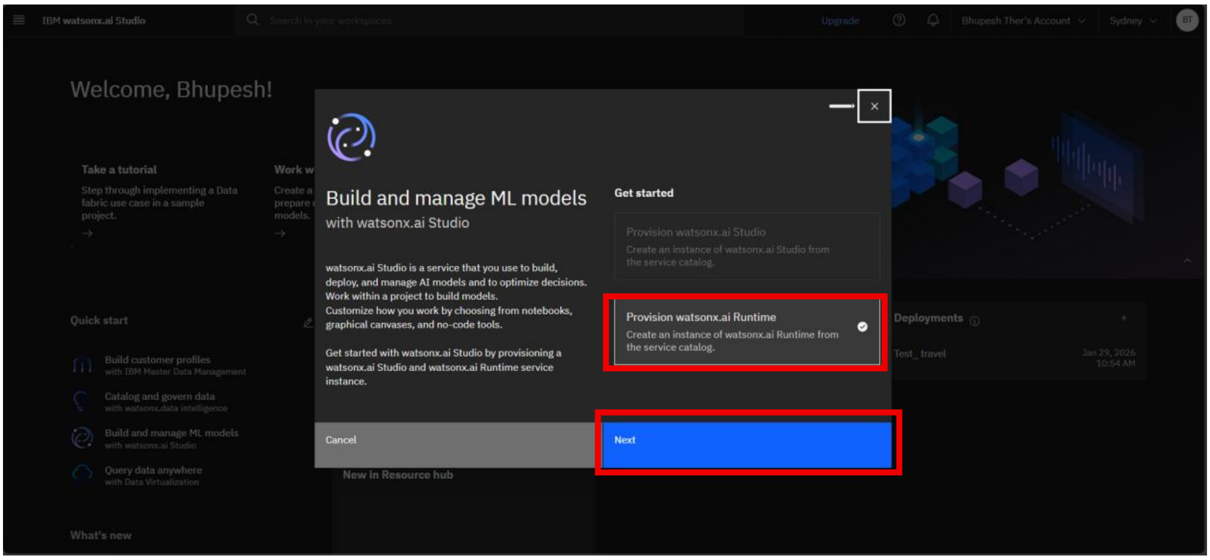
Step-3 create watsonx service select location as Sydney go with free plan and mark checkbox and create.



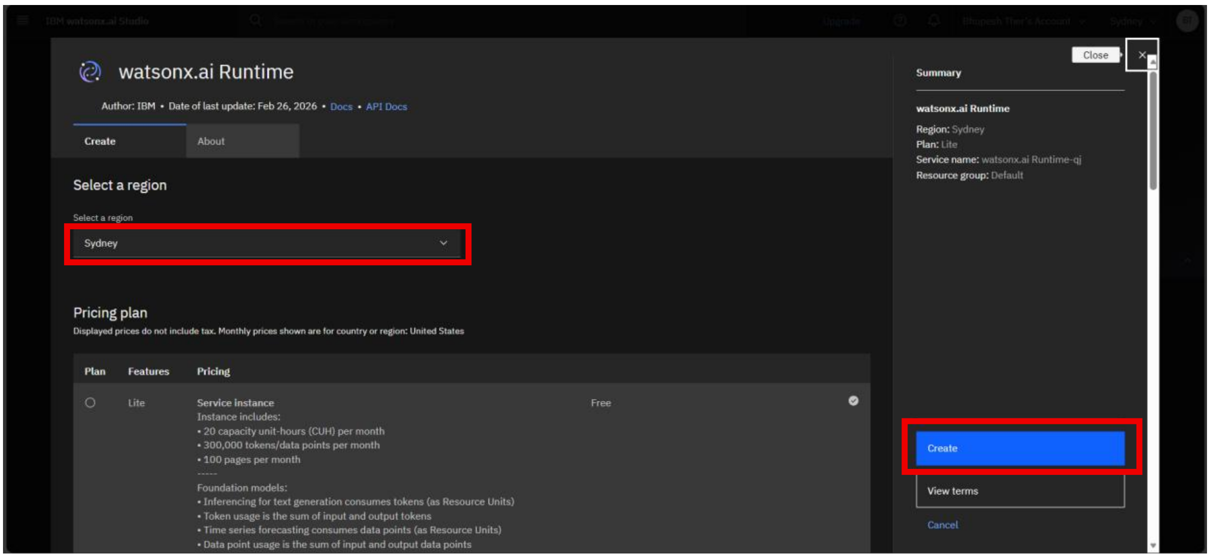
Step-4 click on **launch in** button



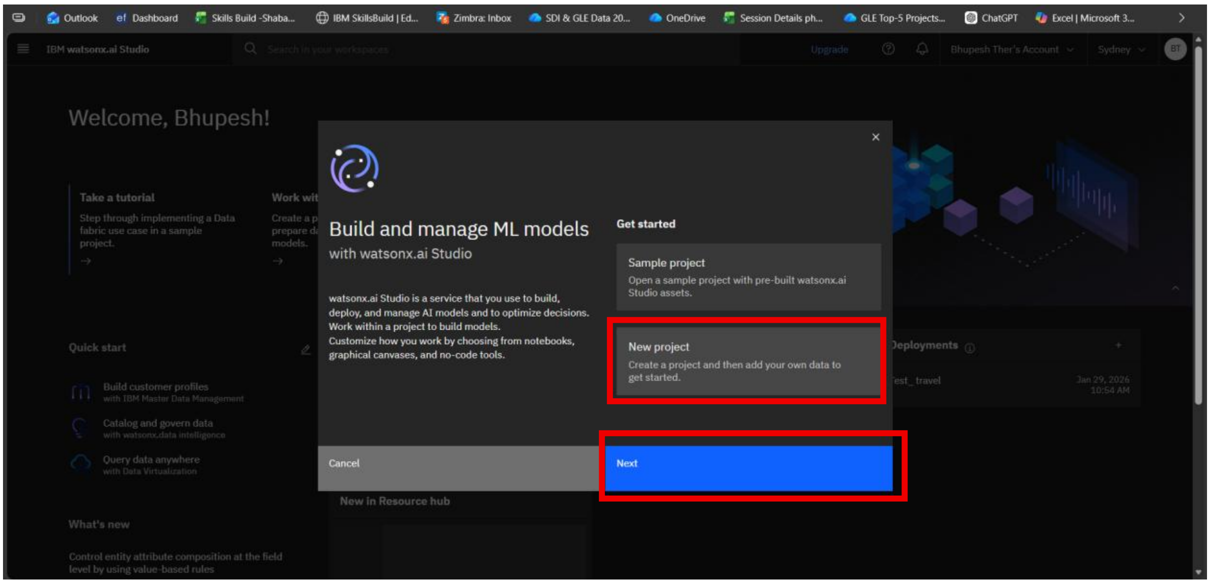
Step-5 Select **watsonx runtime** and click on next.



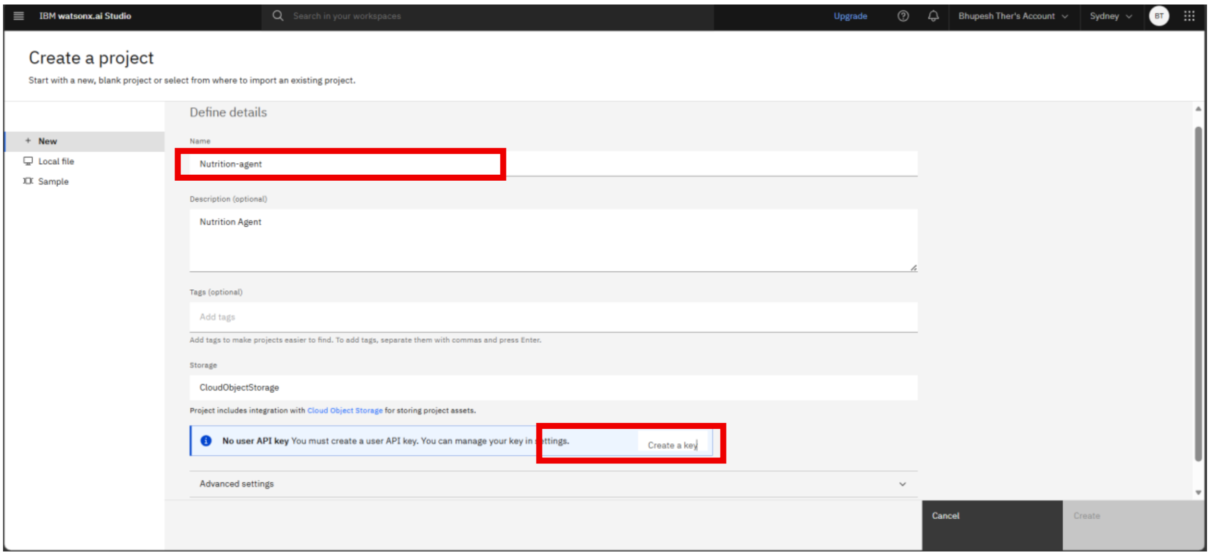
Step-6 select location and click on **create**



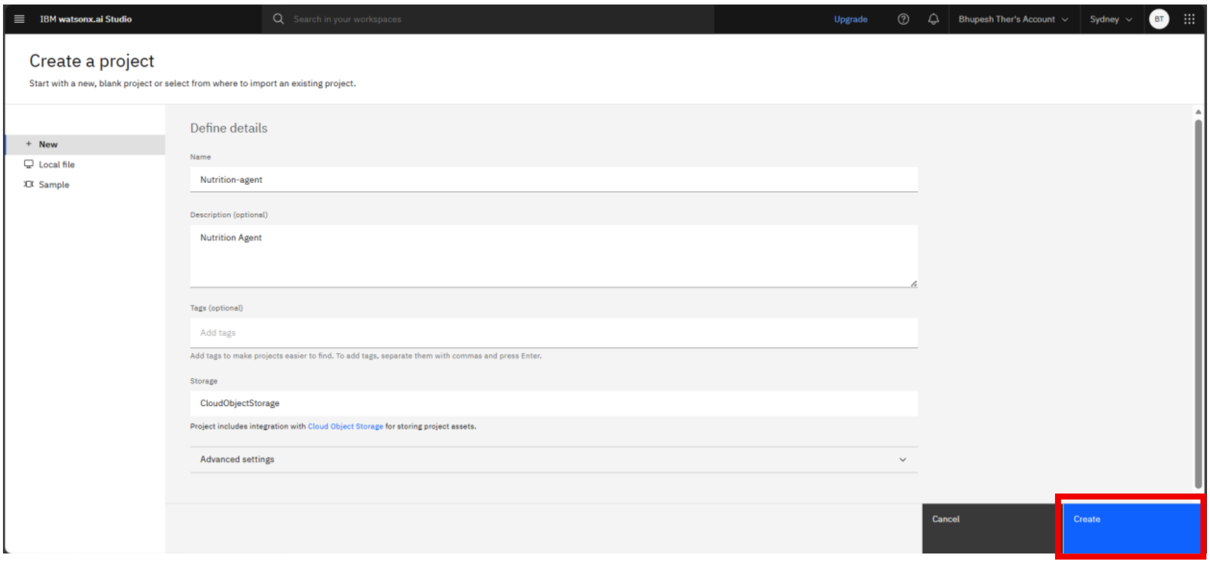
Step-7 Select **New project** and click on **next**.



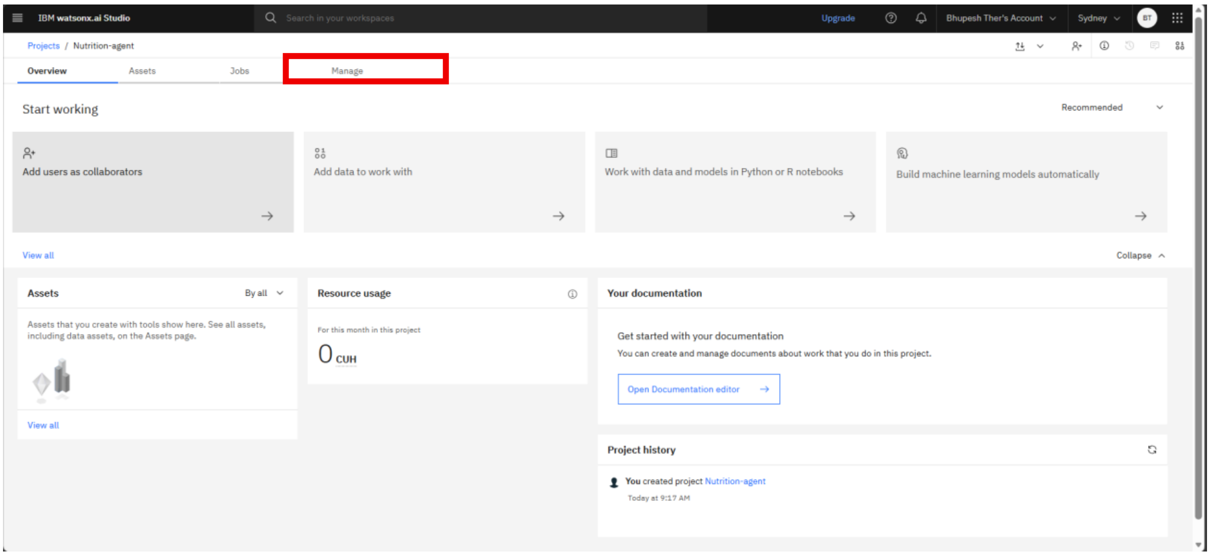
Step-8 give name to your project and click on **Create key**.



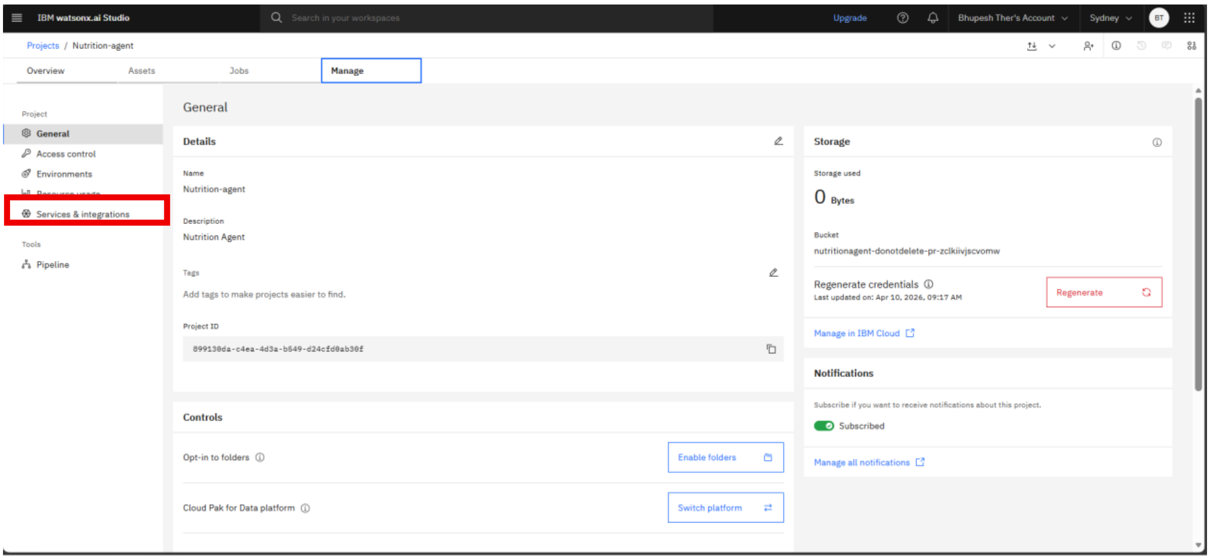
Step-9 click on **create**



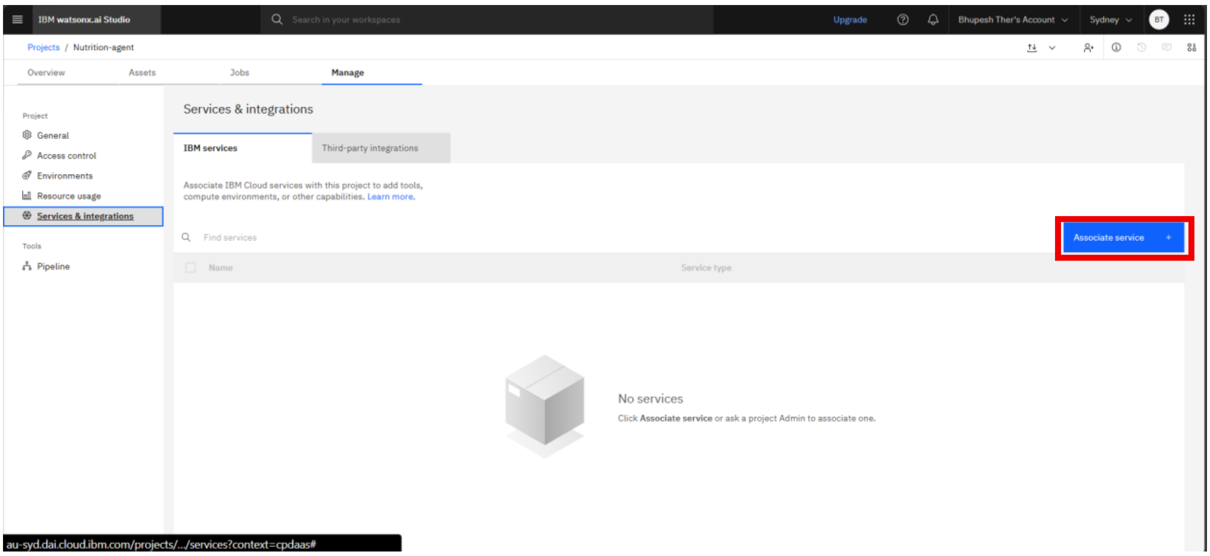
Step-10 go to **manage**



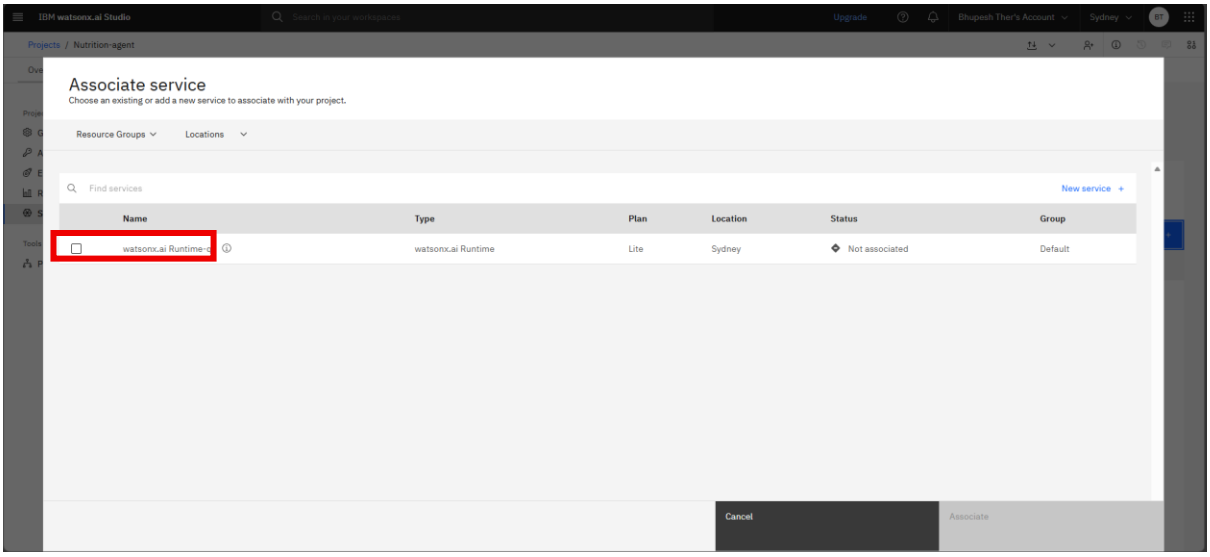
Step-11 click on **services & integration.**



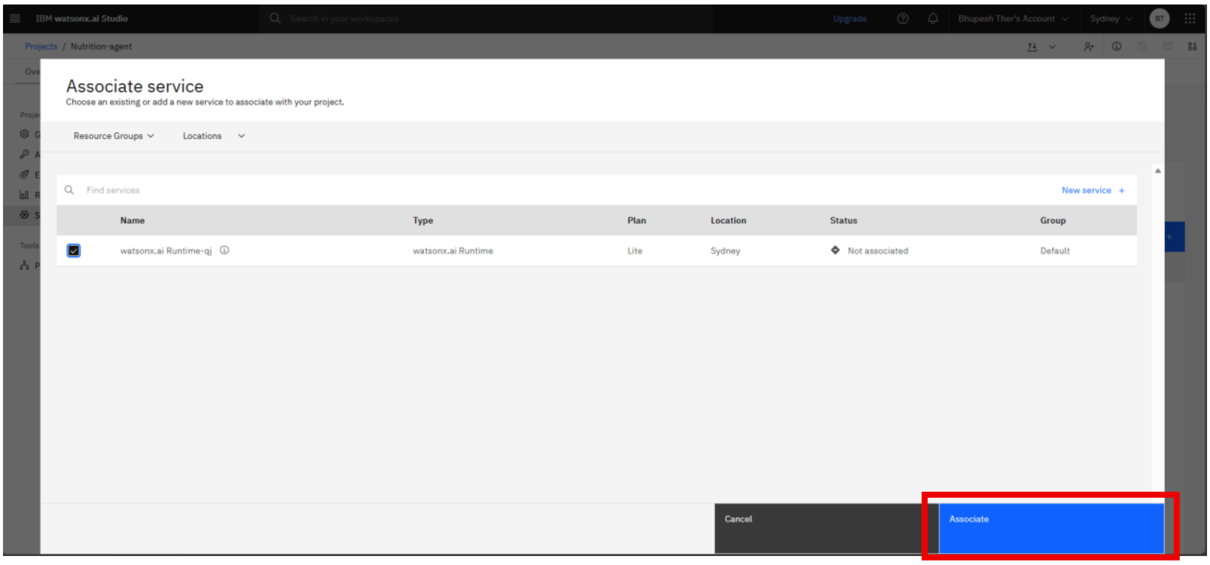
Step-12 click on **Associate Service**.



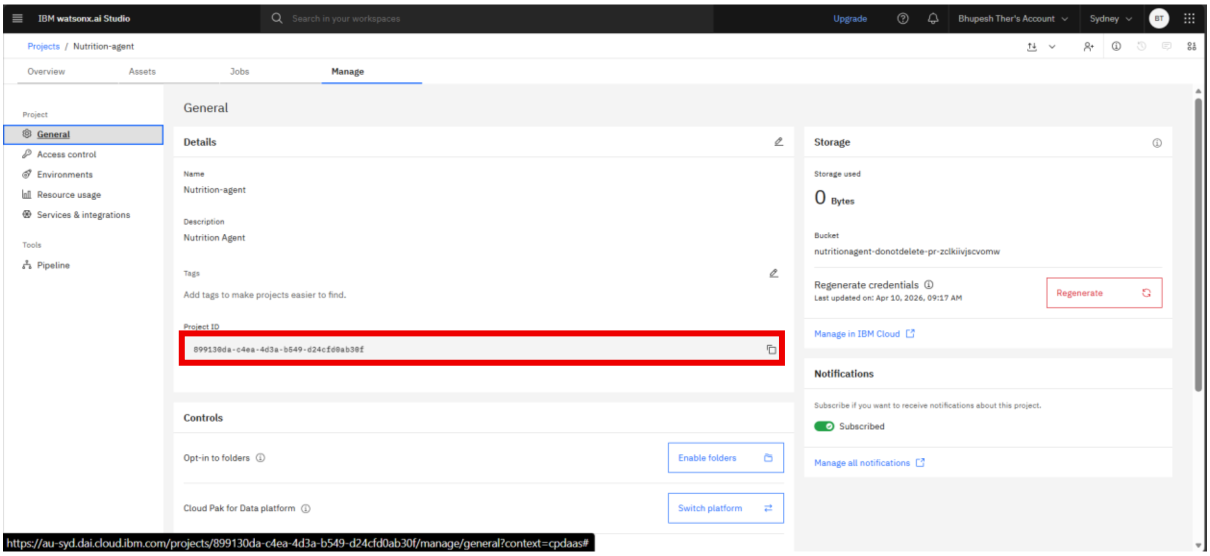
Step-13 mark the **checkbox**.



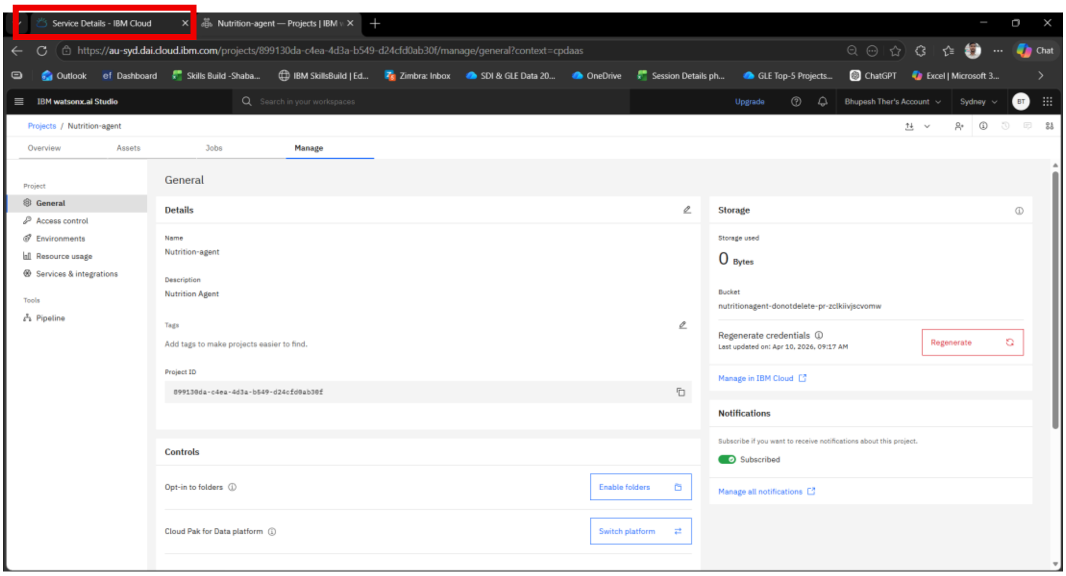
Step-14 click on **create**.



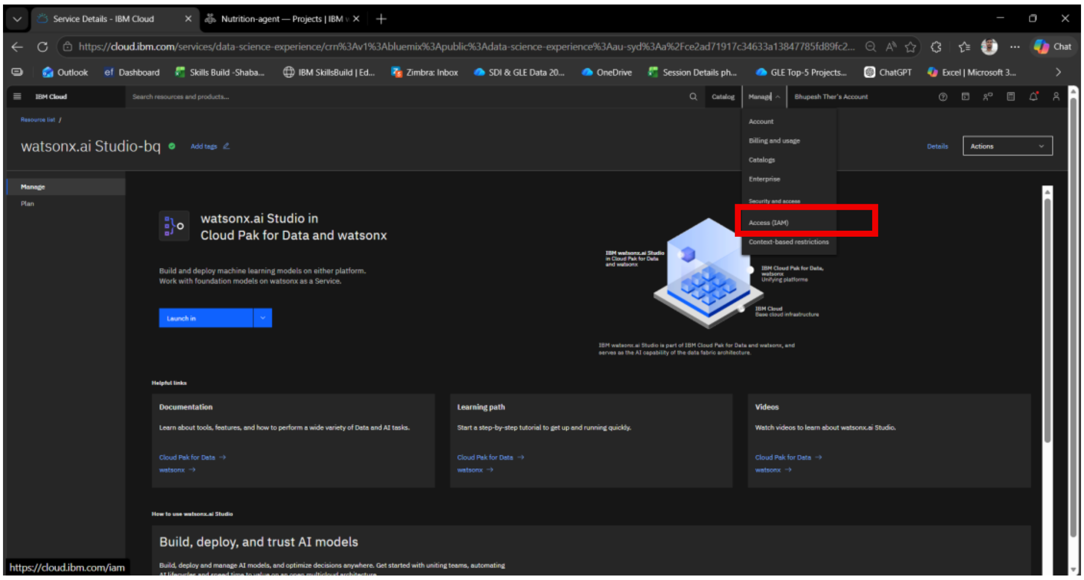
Step-15 Come to general and copy **project ID** and paste in your notepad.



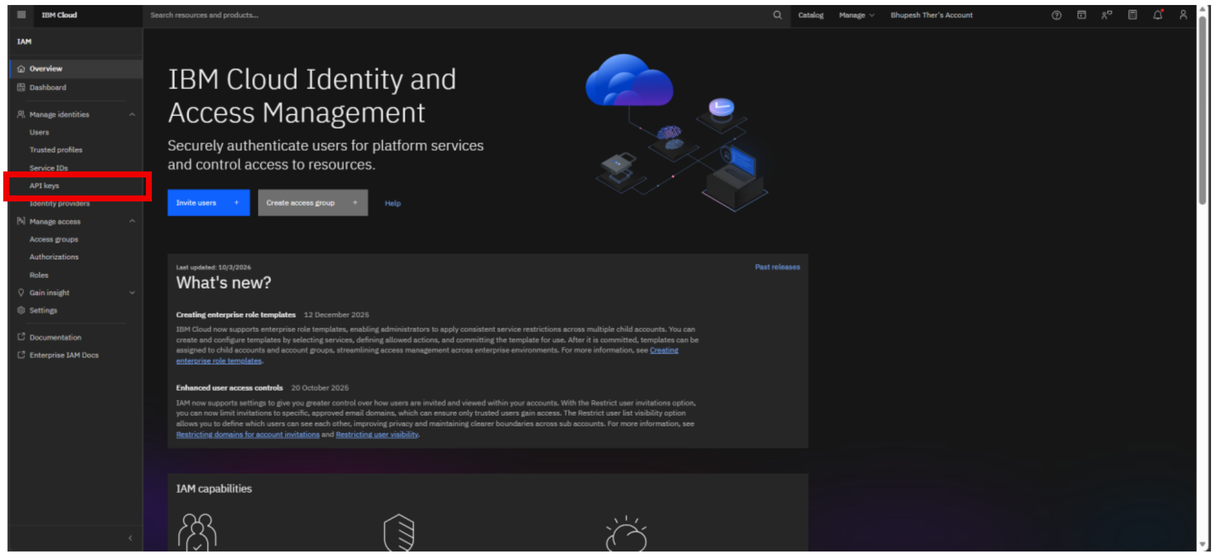
Step-16 Click on previous tab



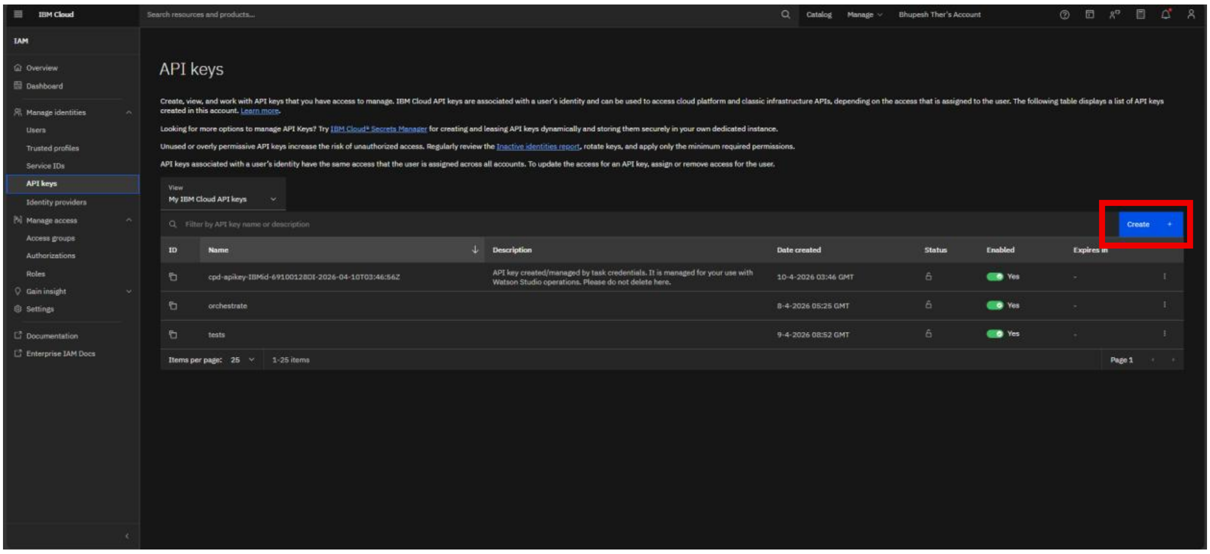
Step-17 click on **Access(IAM).**



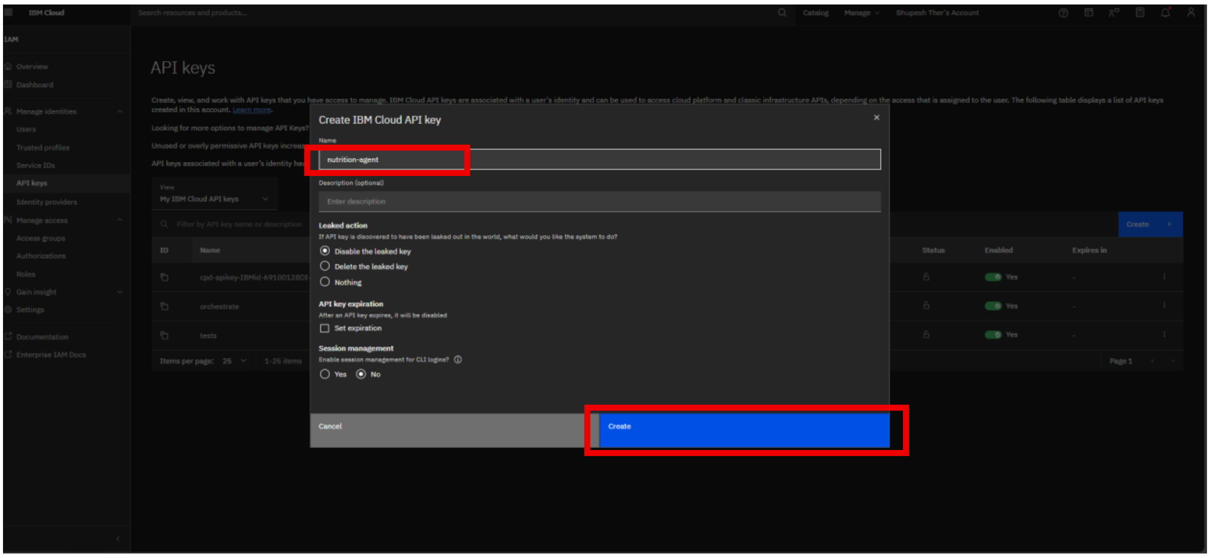
Step-18 click on **API keys**



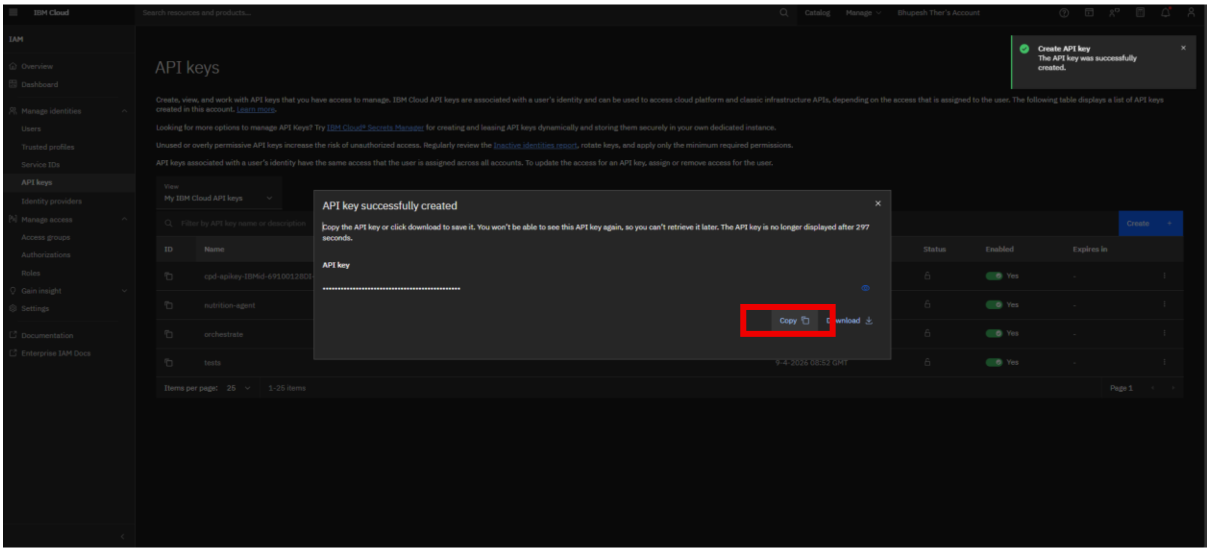
Step-19 click on **create** button



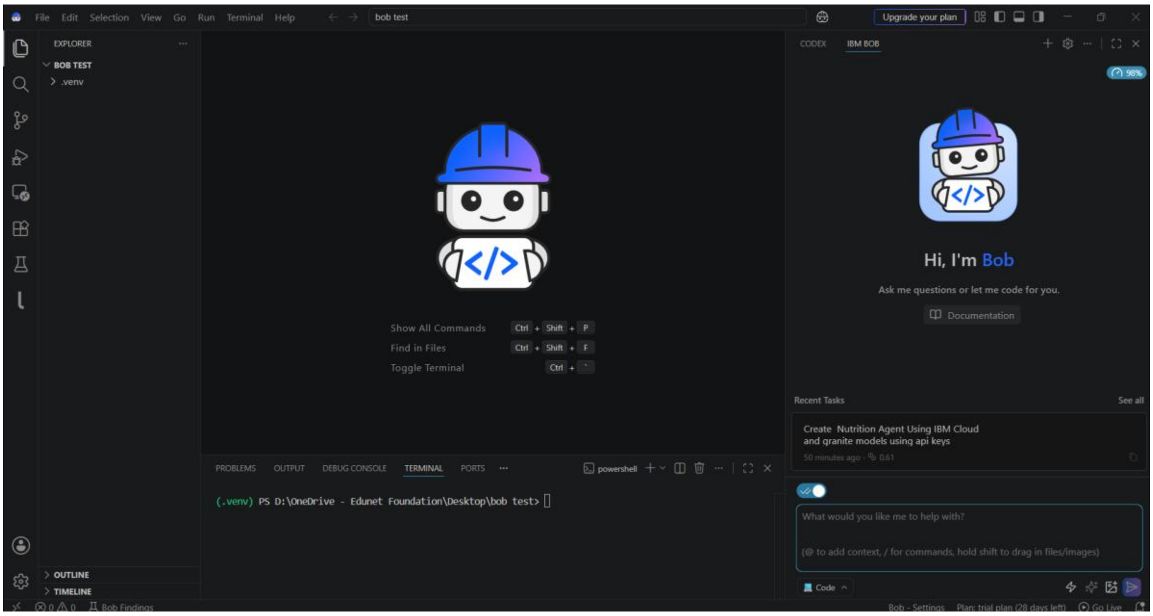
Step-20 give name to your api-key and click on **create**.



Step-21 Copy your **api-key** and paste in your notepad.

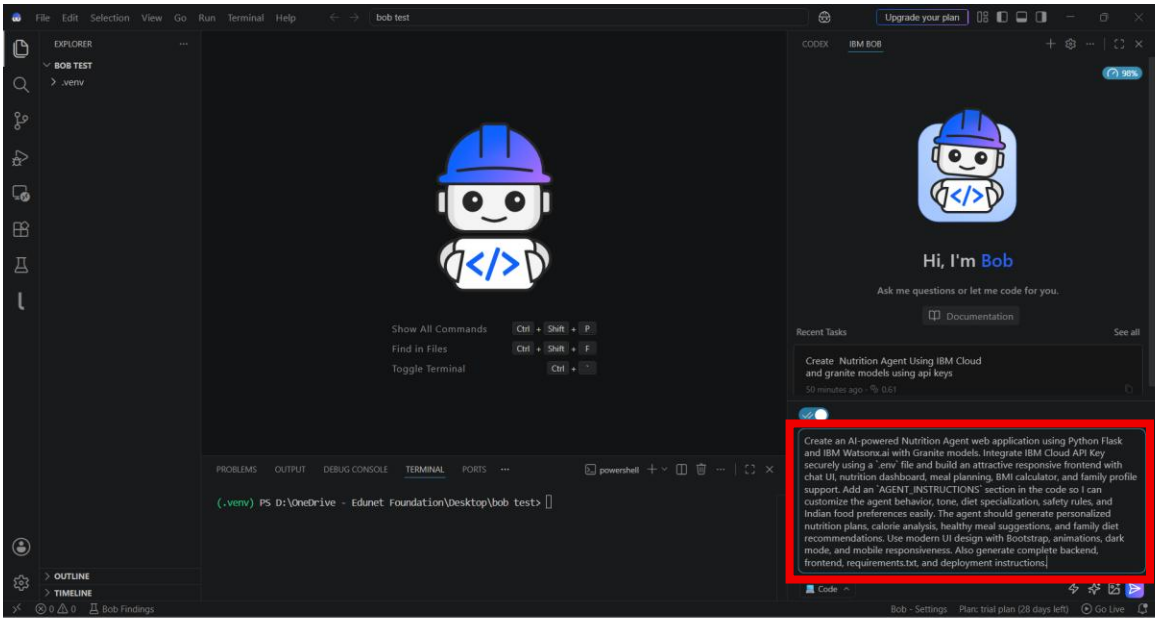


Step-22 Open IBM BOB.

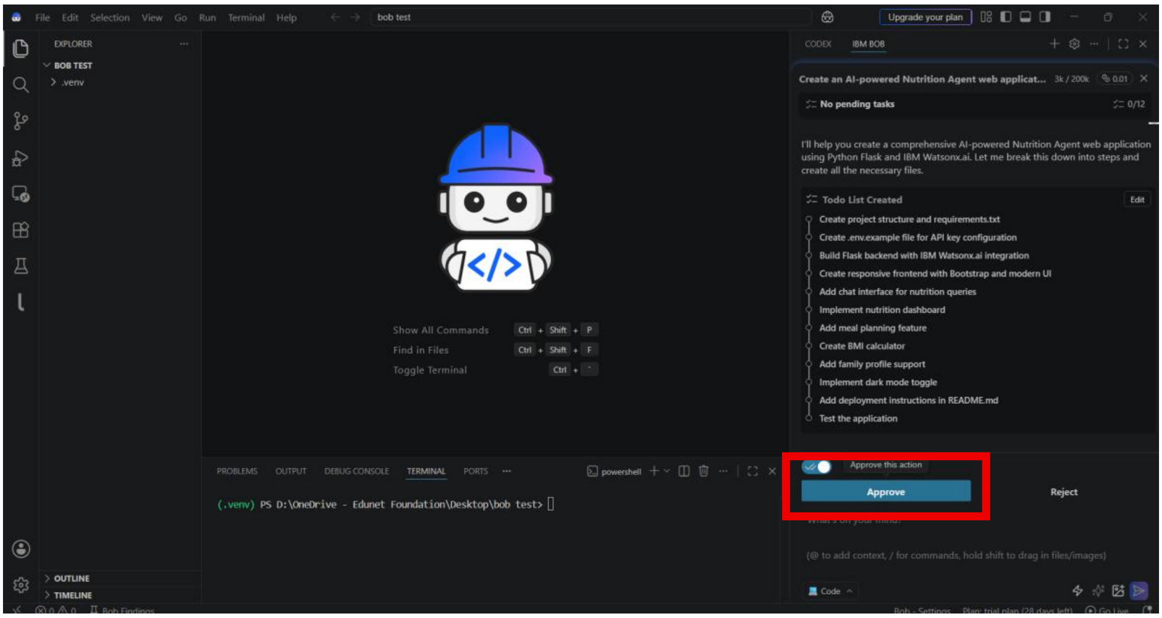


Step-23 Give Prompt to your agent.

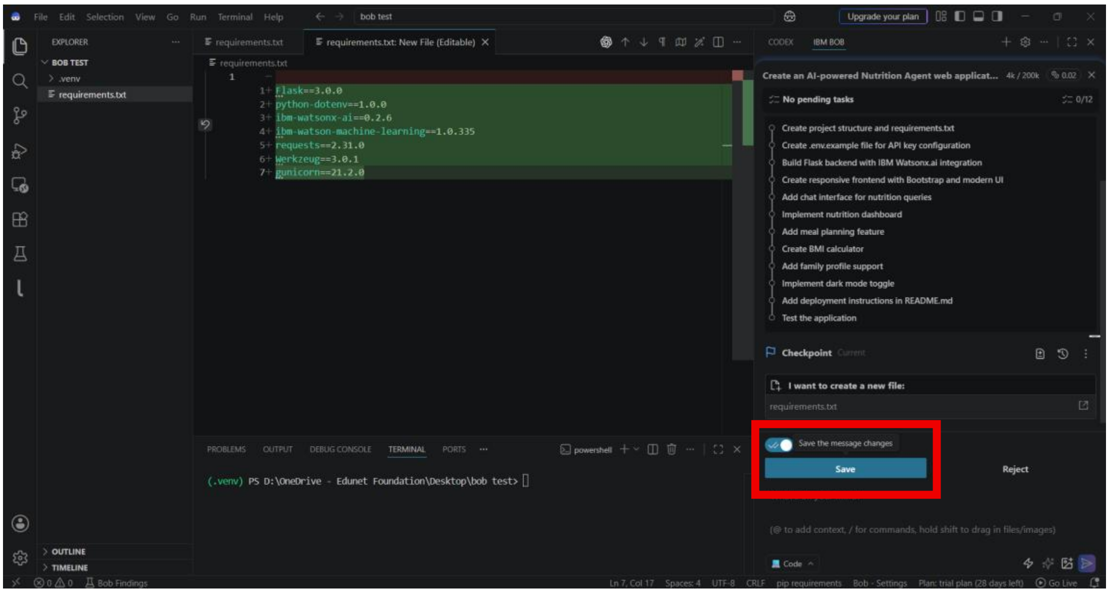
Prompt-
"Create an AI-powered Nutrition Agent web application using Python Flask and IBM Watsonx.ai with Granite models. Integrate IBM Cloud API Key securely using a `.env` file and build an attractive responsive frontend with chat UI, nutrition dashboard, meal planning, BMI calculator, and family profile support. Add an `AGENT_INSTRUCTIONS` section in the code so I can customize the agent behavior, tone, diet specialization, safety rules, and Indian food preferences easily. The agent should generate personalized nutrition plans, calorie analysis, healthy meal suggestions, and family diet recommendations. Use modern UI design with Bootstrap, animations, dark mode, and mobile responsiveness. Also generate complete backend, frontend, requirements.txt, and deployment instructions."



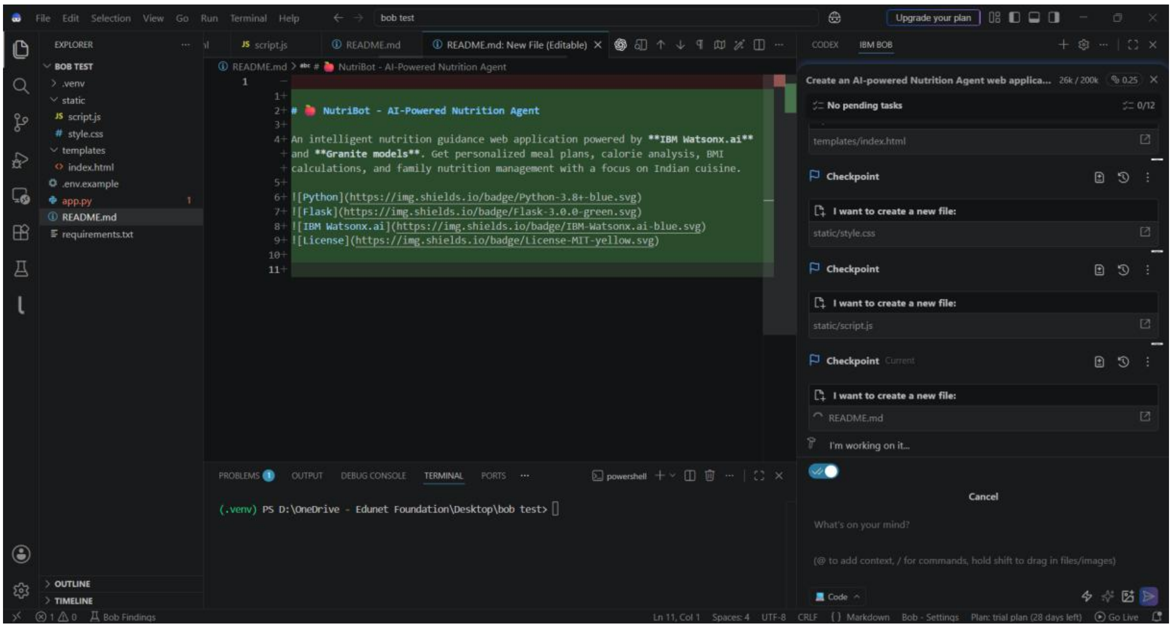
Step-24 Approve the plan created by BOB Agent



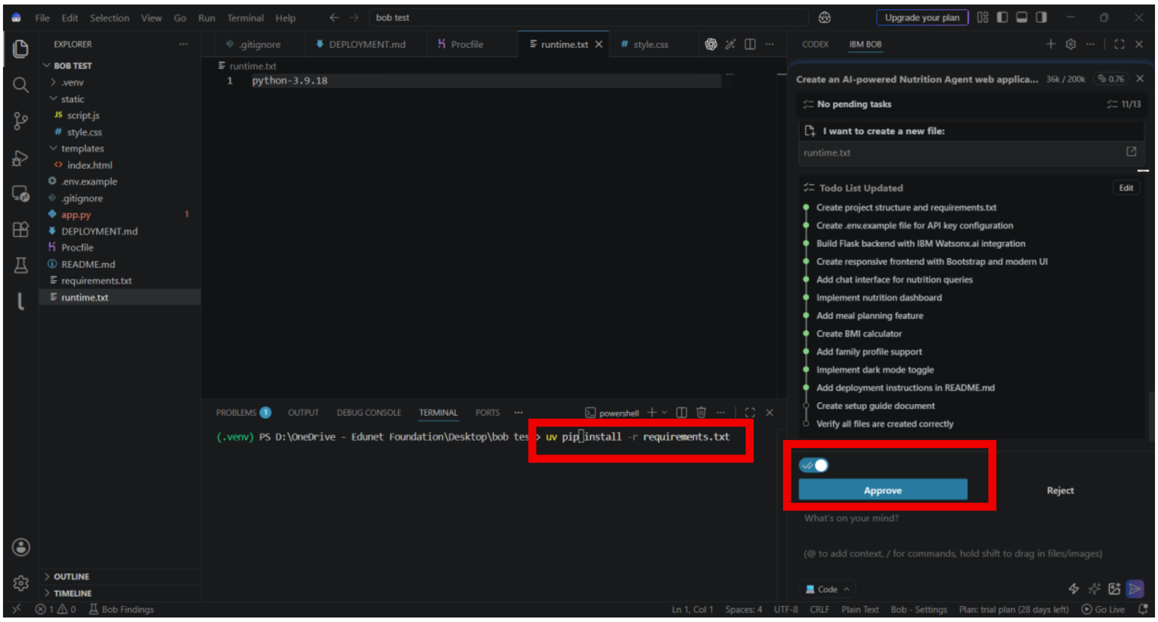
Step-25 Save the code give by BOB.



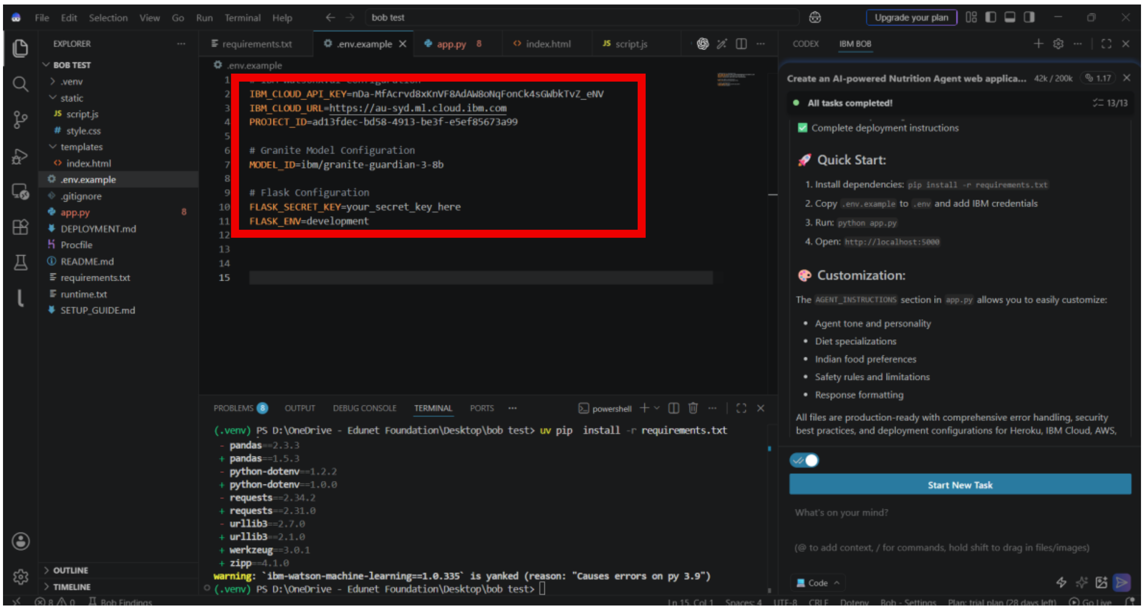
Step-26 It will create code form the scratch.



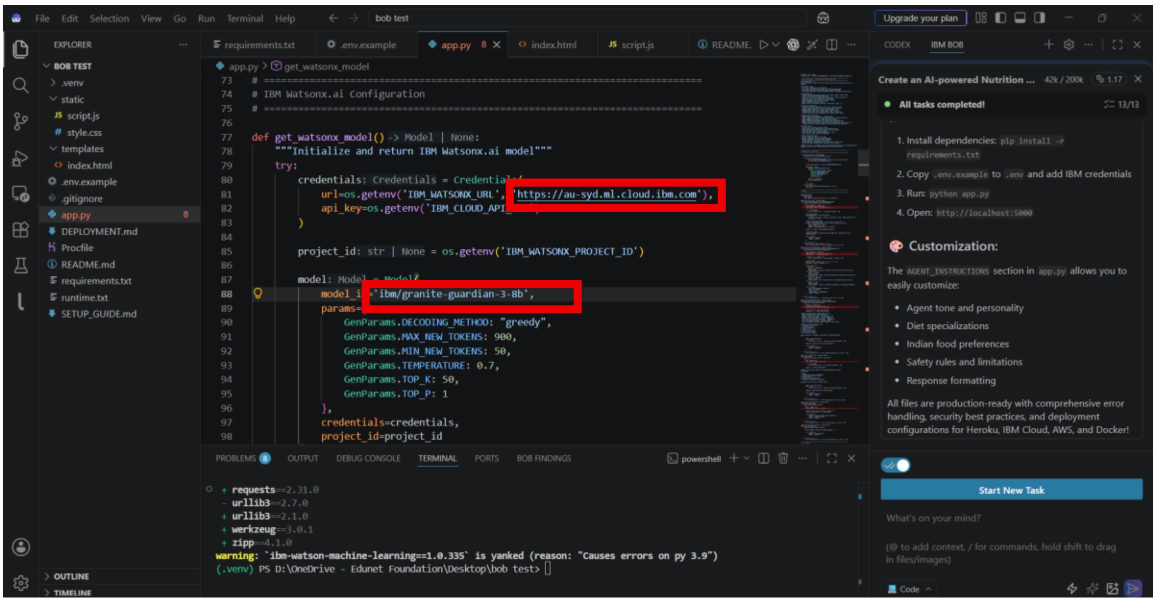
Step-27 once All file created approve for guideline and Now install All libraries in your system using UV.



Step-28 Add **your API Key ,Project ID , IBM URL** and **Model Name in .ENV** file.

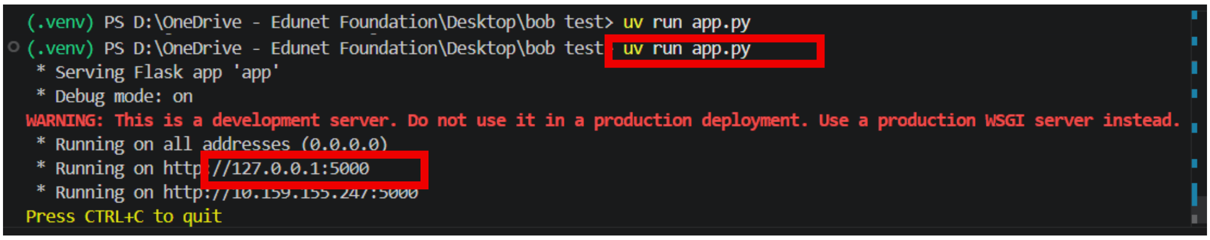


Step-29 Change **URL** Link and Model name From **App.py** .

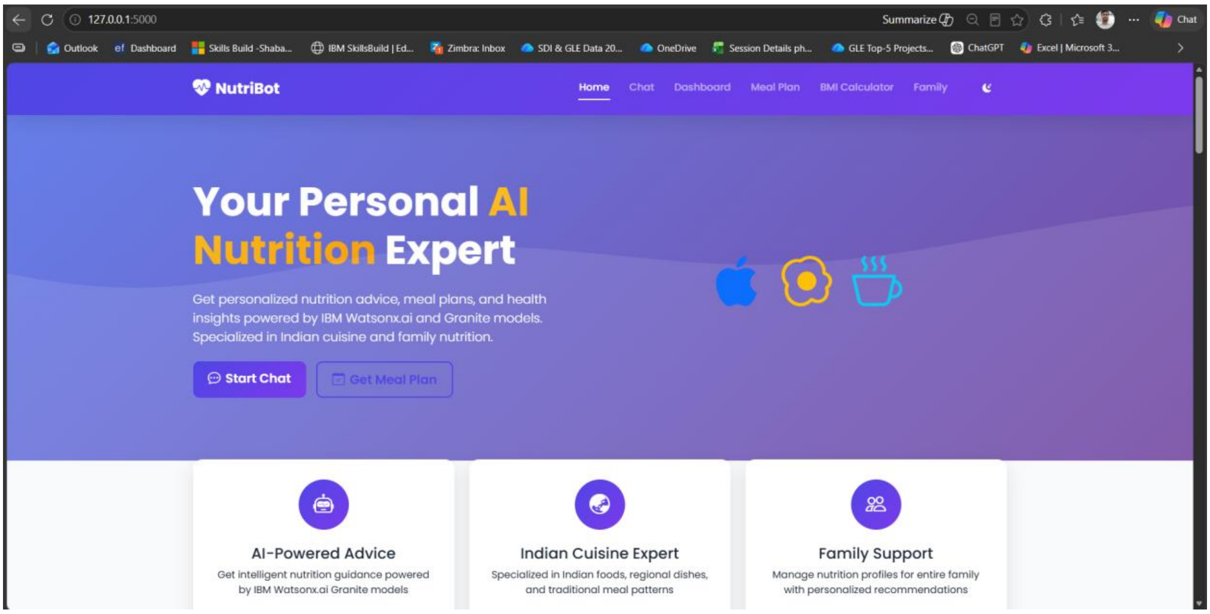


Step-30 Run this Project Using Below Link And Click on Local Host URL

UV run app.py



Step-31 check Website create by BOB.



Step-32 Give Prompt and test your Agent

"Create one day diet plan for me

name :- rajesh

age :- 25

height :- 5.8

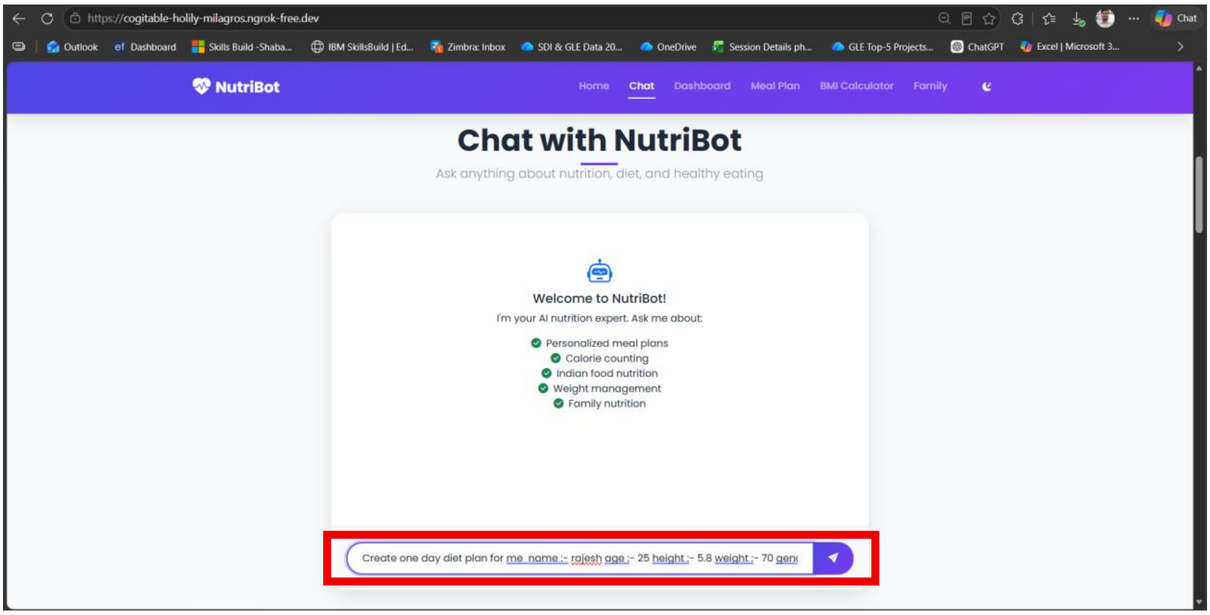
weight :- 70

gender :- male

goal :- weight loss

cuisine preference :- indian

food preferences :- non-vegetarian"



Step-33 Check the Output.